

Key Findings: Geometry of the Virtue of Complexity

5-year rolling backtest · 35 stocks · 6 simulation runs · mapped to the Geometry of Complexity paper

Background

The IPCA model relates returns $r_t \in \mathbb{R}^N$ to characteristics through

$$r_t = Z_t \Gamma^\top f_t + \varepsilon_t, \quad \Gamma \in \text{Mat}_{k,m}(\mathbb{R}), \quad f_t \in \mathbb{R}^k.$$

Γ is identified only through its row space $W = \text{row}(\Gamma) = \text{span}(\Gamma^\top) \in \text{Gr}(k, m)$. After profiling out factors, the objective is a projection functional over the Grassmannian:

$$L(W) = \frac{1}{T} \sum_{t=1}^T \|(I - \Pi_{V_t})r_t\|^2, \quad V_t = \text{span}(Z_t \Gamma^\top) \in \text{Gr}(k, N).$$

With RFF lifting, Z_t is mapped to $\Phi(Z_t) \in \mathbb{R}^{N \times P}$, expanding the admissible subspace family without changing rank k . The estimator uses Riemannian conjugate gradient on $\text{Gr}(m, k)$ with warm-starting.

Diagnostic metrics

All metrics are computed from the fitted $W_t \in \text{Gr}(k, m)$ and the in-window factor series $\hat{f}$.

Sharpe ratio

Performance anchor — Section 7 criterion (iv)

$$\text{SR} = \frac{\text{mean}(r_{\text{port},t})}{\text{std}(r_{\text{port},t})} \times \sqrt{12}, \quad r_{\text{port},t} = \frac{1}{N} \sum_i \text{pos}_{i,t} r_{i,t}, \quad \text{pos}_{i,t} = \hat{r}_{i,t} / \frac{1}{N} \sum_i |\hat{r}_{i,t}|$$

$r_t \in \mathbb{R}^N$: cross-sectional returns (paper Eq. 1). $\hat{r}_{i,t} = \Pi_{V_t} r_t$: projected return prediction.

Subspace stability d_{proj}

Section 4.2, Eq. 17–18

$$d_{\text{proj}}(t) = \|\Pi_{V_t} - \Pi_{V_{t-1}}\|_F = \sqrt{2 \sum_{i=1}^k \sin^2 \theta_i}, \quad \max = \sqrt{2k}$$

$\Pi_{V_t} = W_t W_t^\top$: projector onto $V_t \in \text{Gr}(k, m)$. $\theta_i = \arccos \sigma_i(W_t^\top W_{t-1})$: k principal angles. *Note: currently confounded by random restarts — see Finding 4.*

Effective rank $\text{erank}(\hat{\Sigma}_f)$

Definition 4.1 + Theorem 6.1(iii)

$$\text{erank}(\hat{\Sigma}_f) = \exp(H(p)), \quad H(p) = - \sum_i p_i \log p_i, \quad p_i = \lambda_i(\hat{\Sigma}_f) / \text{tr}(\hat{\Sigma}_f)$$

$\hat{\Sigma}_f = \frac{1}{T} \hat{f}^\top \hat{f} \in \mathbb{R}^{k \times k}$. $\hat{f}_t = (\Lambda_t^\top \Lambda_t)^{-1} \Lambda_t^\top r_t \in \mathbb{R}^k$, $\Lambda_t = Z_t W_t \in \mathbb{R}^{N \times k}$. Theorem 6.1(iii): $\text{spec}(\Sigma_f^*) = \{\lambda_1, \dots, \lambda_{k_0}, \underbrace{\sigma^2, \dots, \sigma^2}_{k-k_0}\}$ collapses $\text{erank} \rightarrow k_0$ when $k > k_0$.

Spectral gap sg

Theorem 6.1(ii) and 6.1(iii)

$$\text{sg} = \lambda_k(\hat{\Sigma}_f) / \lambda_1(\hat{\Sigma}_f) \approx (\lambda_k - \sigma^2) / \lambda_1$$

Proxy for the exact gap in Theorem 6.1(iii); exact computation requires fitting $k+1$ factors. Theorem 6.1(ii): $\text{Hess}_V J[\xi, \xi] \propto \sum_{i=1}^{k_0} (\lambda_i - \sigma^2) \|\xi_i\|^2$; regularisation raises $\lambda_i - \sigma^2$, increasing curvature and the spectral gap.

Finding 1 — Effective rank reveals structural complexity limits

Geometry of Complexity — Definition 4.1 + Theorem 6.1(iii). The intrinsic dimension of a return representation is $\text{erank}(\hat{\Sigma}_f)$. Theorem 6.1(iii) proves $\text{spec}(\Sigma_f^*)$ has block structure: k_0 signal eigenvalues above σ^2 , then $k - k_0$ eigenvalues exactly equal to σ^2 . This collapses erank toward k_0 and creates a growing gap below the $\text{erank} = k$ diagonal.

The growing gap from the diagonal identifies k_0

Plotting erank against k alongside the $\text{erank} = k$ diagonal reveals a gap that widens progressively with k . Each factor dimension beyond k_0 adds one eigenvalue equal to σ^2 , pulling erank downward. The gap steepens between $k = 8$ and $k = 12$, consistent with $k_0 \approx 8$.

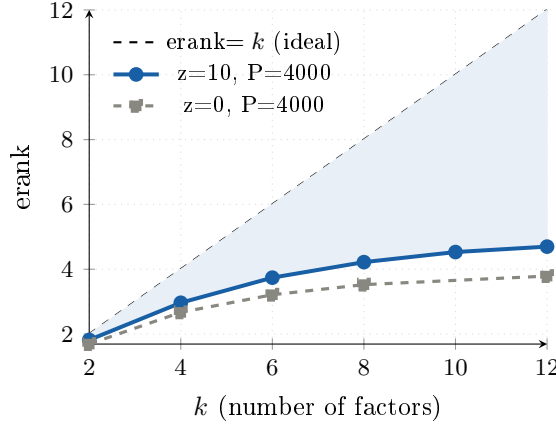


Figure 1: erank (solid/dashed) vs the $\text{erank} = k$ diagonal (black dashed). Shaded area = noise contamination predicted by Theorem 6.1(iii). Gap widens with k ; steepening near $k = 8$ – 12 marks k_0 .

erank peaks at $P \approx 1000$ – 4000 — parameter vs structural complexity

At fixed $k = 4$, $z = 10$, erank rises from 3.04 at $P = 64$ to a peak near $P = 256$ – 1000 , then declines to 2.90 at $P = 12,000$. This non-monotone behaviour maps to Section 3.5: RFF lifting enlarges the admissible subspace family (parameter complexity) but additional flexibility is structurally meaningful only if it yields persistent increases in intrinsic dimension. Beyond $P \approx 1000$ the RFF approximation introduces more noise directions than signal, and erank declines. The spectral gap does not show this P -dependence — confirming the paper’s separation between parameter and structural complexity.

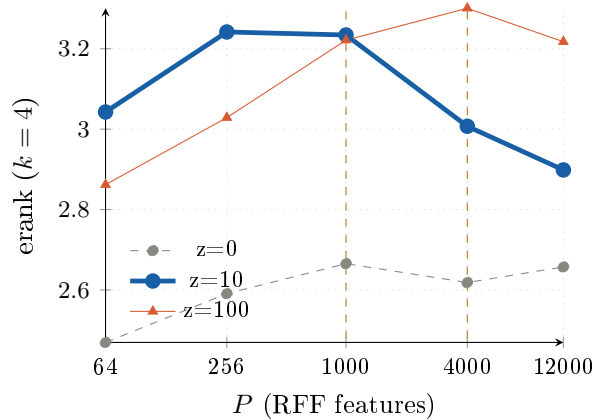


Figure 2: erank vs P at $k = 4$ for three regularisation values. The non-monotone shape: erank peaks then declines beyond $P \approx 1000$ – 4000 . The spectral gap does not show this behaviour, confirming the Section 3.5 parameter/structural complexity separation.

Finding 2 — Spectral gap: the strongest diagnostic

Geometry of Complexity — Theorem 6.1(ii) and 6.1(iii). Theorem 6.1(iii): $\text{spec}(\Sigma_f^*) = \{\lambda_1, \dots, \lambda_{k_0}, \sigma^2, \dots, \sigma^2\}$. The spectral gap $\text{sg} = \lambda_k/\lambda_1$ collapses toward σ^2/λ_1 as k exceeds k_0 . Theorem 6.1(ii): $\text{Hess}J \propto \sum(\lambda_i - \sigma^2)\|\xi_i\|^2$; regularisation raises eigenvalue gaps, increasing curvature.

In the joint (k, z) grid (Run 6, 30 configurations), the spectral gap is monotonically decreasing in k and monotonically increasing in z — no exceptions. The key structural feature is the **98% relative collapse from $k = 8$ to $k = 12$** , present at every z value (96–99%). All earlier steps show 60–77% drops. This structural break is independent of regularisation, identifying $k_0 \approx 8$. At $k = 12$ the gap is essentially zero at every z (0.0001–0.0005). Critically, the spectral gap is P-independent (CV < 7% across all P values).

	z=0	z=1	z=5	z=10	z=50	z=100	
k=2	0.355	0.446	0.467	0.470	0.504	0.544	
k=4	0.080	0.108	0.122	0.121	0.150	0.213	
k=6	0.019	0.034	0.037	0.040	0.055	0.086	
k=8	0.005	0.011	0.012	0.013	0.019	0.031	
k=12	0.000	0.000	0.000	0.000	0.000	0.000	98% collapse

factors k

regularisation z

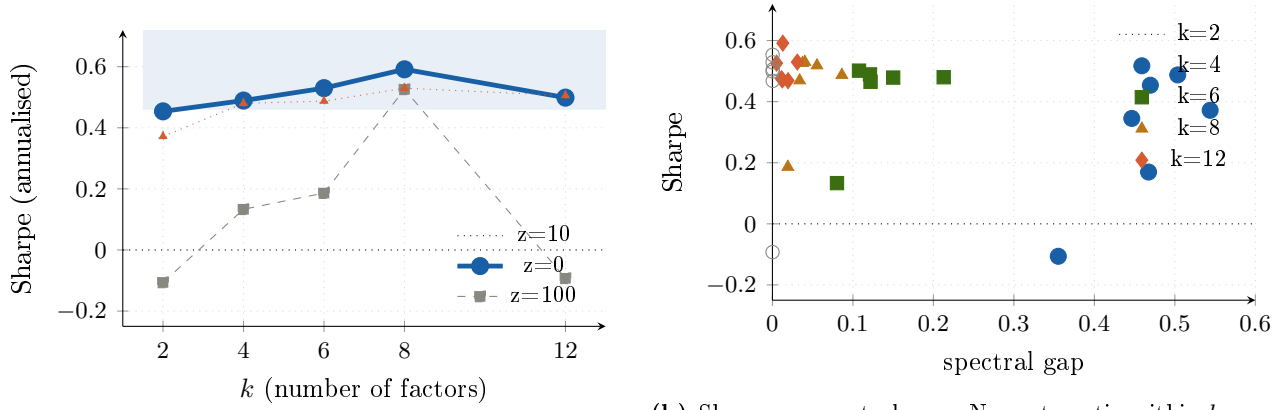
Figure 3: Spectral gap ratio heatmap. Monotone in both dimensions without exception. Coral line marks the $k = 8 \rightarrow 12$ boundary where 98% structural collapse occurs at every z . The $k = 12$ row is near-zero for all z .

Finding 3 — Sharpe: virtuous arc only visible with regularisation

With only 60 monthly portfolio return observations, the standard error on an annualised Sharpe of ≈ 0.5 is approximately 0.13. Differences smaller than this are statistically indistinguishable. Read these plots for qualitative pattern rather than precise values.

At $z = 0$ (illusory regime) Sharpe is flat and low across all k — no k selection helps. At $z = 10$ a clear arc emerges, peaking at $k = 8$ (Sharpe = 0.591) and declining at $k = 12$. At $z = 100$ a similar but lower arc appears. This maps to Section 7: complexity is virtuous if out-of-sample performance improves with k . Here it only improves when the model is in the virtuous regime ($z \geq 10$). The peak at $k = 8$ corroborates $k_0 \approx 8$ identified from the spectral gap.

The scatter plot below shows Sharpe against the spectral gap for all 30 (k, z) cells. Within each k , higher spectral gap does not reliably predict higher Sharpe. This is consistent with the paper: the spectral gap diagnoses factor identification quality (Section 7 criterion ii–iii) while Sharpe diagnoses performance (criterion iv). Both are necessary; they are complementary, not redundant.



(a) Sharpe vs k . Shaded band = ± 1 SE at $k = 8$. Only $z = 10$ shows a virtuous arc.

(b) Sharpe vs spectral gap. No systematic within- k relationship: diagnostics and performance are complementary.

Figure 4: Sharpe plots (Run 6, $P = 4000$). Left: virtuous arc appears only at $z \geq 10$. Right: spectral gap predicts factor structure quality, not performance directly.

Summary mapping

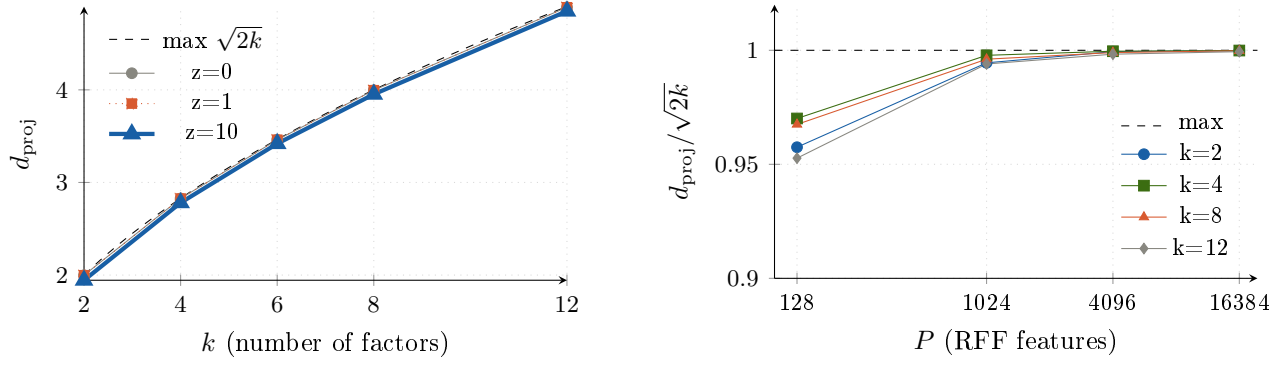
Finding	Theorem / Section	Evidence	Strength
erank gap from diagonal grows with k	Thm 6.1(iii) block spectral structure	Gap tracks $k - k_0$ noise dims	Strong
erank peaks at $P \sim 1000-4000$	Section 3.5 param vs structural	Non-monotone in P at every z	Moderate
Spectral gap 98% collapse at $k = 8 \rightarrow 12$	Thm 6.1(iii) block spectral structure	98% drop at all 6 z values, P -independent	Strong — $k_0 \approx 8$
Spectral gap monotone in k and z	Thm 6.1(ii)+(iii)	30/30 cells, no exceptions	Strong
Sharpe arc at $z = 10$, flat at $z = 0$	Section 7 virtuous vs illusory	Peak 0.591 at $k = 8$; SE ≈ 0.13	Moderate, directional
d_{proj} saturates at $\sqrt{2k}$	Cannot confirm Thm 6.1(i)	Matches Haar-random formula to 0.01%	Unreliable: warm-start needed

Finding 4 — d_{proj} is unreliable: random restarts explain the saturation

Why d_{proj} cannot confirm Theorem 6.1(i) in the current setup. Theorem 6.1(i) predicts flat Hessian directions produce $O(1)$ subspace rotations. However, the backtest uses a fresh random W_0 at every rolling window. Two points drawn from the Haar-uniform measure on $\text{Gr}(m, k)$ have $\mathbb{E}[d_{\text{proj}}^2] = 2k(m - k)/(m + 1)$, which approaches $2k$ as $m = P$ grows — regardless of loss curvature. The flat Hessian and random restarts produce identical d_{proj} signatures and cannot be distinguished without warm-starting.

At $m = P = 4000$, $k = 6$: the Haar formula gives $\mathbb{E}[d_{\text{proj}}] = 3.461$ and we observe 3.461. The ratio observed/expected_{random} = 1.0001. This holds at every $(k, P, z = 0)$ configuration tested. One partial observation survives: at small $P = 64$, $z = 10$ gives $d_{\text{proj}} = 1.51$ (53% of max) vs $z = 0$ at 1.95 (98% of max), a genuine gap since the Haar baseline at $P = 64$ is only 97.7% of max. **The fix**

is one line: pass `initial_point=W_prev` to `est.fit()`.



(a) d_{proj} vs k . All z values near max at $P = 4000$; $z=0$ and $z=1$ indistinguishable.

(b) $d_{\text{proj}}/\sqrt{2k}$ vs P at $z = 0$. All k converge to 1.0 as P grows — matching the Haar-random expectation.

Figure 5: d_{proj} plots (Run 6 left, Run 2 right). Saturation is driven by ambient Grassmannian geometry, not loss curvature.

Open questions and caveats

Warm-start — critical fix required. The fix is one line in the rolling loop: `initial_point=W_prev`. Until done, all d_{proj} -based conclusions are provisional.

Kelly mapping incomplete. Both Kelly papers (2024 *Journal of Finance*; 2025 SSRN) are cited by the Geometry of Complexity paper but could not be read directly. The Sharpe-based complexity scaling test requires $N \gg 35$ to have statistical power.

k_0 is estimated, not formally identified. The spectral gap 98% collapse at $k = 8 \rightarrow 12$ and Sharpe peak at $k = 8$ suggest $k_0 \approx 8$ for this 35-stock universe. A formal test using Theorem 5.1 (fitting at $k = k_0$ should yield a unique minimiser) has not been implemented.